

Elanchezhian Muthukumar

Software Developer

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SKILLS

- **Languages:** Python, JavaScript (ES6+)
- **Frontend:** React.js, Next.js, HTML5, CSS3, Bootstrap, Framer Motion
- **Backend & APIs:** FastAPI, Django, Flask, RESTful APIs, Google Drive API, Gemini AI API
- **Databases & Cloud:** MySQL, PostgreSQL, MongoDB, Supabase, AWS, Google Cloud Console
- **Tools & DevOps:** Git, GitHub, Figma, Visily AI, Postman
- **Familiar with:** Docker (Beginner)

EDUCATION

- **PERIYAR UNIVERSITY**
MASTER OF COMPUTER APPLICATIONS — University Rank Holder
Aug '22 — Jun '24
CGPA: 8.5
- **VYSYA COLLEGE**
BACHELOR OF SCIENCE IN COMPUTER SCIENCE
Jul '19 — Jun '22
CGPA: 7.7

CERTIFICATIONS

- **Full Stack Web Developer Certified** – Udemy (Jun '24).
- **Java and Python Programming Certification** – NPTEL (Feb '23).
- **AI for India 2.0 Certification** – GUIV, Skill India Digital (Aug '24).
- **Innovation, Design, and Entrepreneurship (IDE) Bootcamp** – MIC and AICTE (Jun '23).

AWARDS AND HONORS

- **1st Place in Project Expo, Emerge 2K23** at SRM Valliammai Engineering College (Nov '23).
- **Consolation Prize in In-House Project Expo-2K23** by Periyar University (Sep '23).

PROFESSIONAL EXPERIENCE

SOFTWARE DEVELOPER TRAINEE

MAHIMA TECHNOLOGY PRIVATE LIMITED | SALEM, TAMIL NADU, INDIA
Jul '24 — Present

- Built a **comprehensive time-tracking system** to monitor task states (WIP, Breaks, Calls, Meetings, etc.) and improve productivity.
- Engineered **role-based dashboards** for admins and TLs, improving task management and boosting efficiency by **20%**.
- Integrated **Gemini AI API** to extract structured invoice data from **PDFs (via S3)** and **Excel files**, reducing manual entry by **30%**.
- Connected with the **Google Drive API** for automated dataset processing, improving workflow speed by **30%**.
- Enhanced scalability with **asynchronous processing**, decreasing large file handling time by **40%**.

WEB DEVELOPMENT INTERN

RAKUMURA IT SOLUTIONS | CHENNAI, TAMIL NADU, INDIA
Jan '24 — Apr '24

- Executed end-to-end website projects, enhancing user experience and optimizing responsive design, which resulted in a **40% increase in client engagement**.
- Orchestrated collaboration with cross-functional teams, including marketing and project management, to develop 10 innovative initiatives, boosting customer engagement by **40%** and **driving a 25%** rise in quarterly sales.

PROJECTS

SECURE AI DOCUMENT & BOOKMARK VAULT

NEXT.JS | GEMINI 2.5 AI | TESSERACT.JS | CRYPTOJS | GOOGLE DRIVE API | 2FA
[\[LINK\]](#)

- **Architected a zero-knowledge security model** using client-side AES encryption (CryptoJS), ensuring all data stored on a user's personal Google Drive remains completely private and unreadable without their Master Password.
- **Engineered a client-side OCR pipeline** (Tesseract.js) with advanced in-browser image pre-processing to automatically extract, parse, and structure data from uploaded documents like IDs, receipts, and certificates.
- **Integrated Gemini 2.5 AI** to enable natural language vault management (voice & text), translating complex user prompts into executable application commands for seamless organization.
- **Implemented a multi-layered security framework** featuring Google OAuth 2.0 with minimal scopes, user-enabled **2-Factor Authentication (2FA)**, and configurable auto-locking inactivity timers.

VIRTUAL ASSISTANT FOR INDIVIDUALS WITH DISABILITIES

HTML5 | CSS3 | JAVASCRIPT | BOOTSTRAP | PYTHON | FLASK | REST APIs
[\[LINK\]](#)

- Designed and developed an **AI-powered desktop assistant** to enhance computer accessibility for users with physical disabilities (blind, deaf, or non-verbal).
- Integrated a **speech-to-text (STT)** engine to enable real-time, voice-based interaction and hands-free communication.
- Implemented **image and speech processing modules** for interpreting user input and facilitating responsive desktop task control.
- Built a modular Flask backend with REST APIs for intelligent task automation and rapid response across assistive features.
- Deployed and tested with **10+ real users**, improving interaction efficiency by over **50%** and empowering independent computer use.